

BevOS: A Centralized Workflow Management Platform

Lab 1 – Draft 3

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1 Introduction

This section introduces the challenges associated with coordinating beverage manufacturing operations and the characteristics needed to address them. It also provides an overview of BevOS and how the proposed system is intended to improve the beverage manufacturing workflow.

1.1 Societal Problem

The beverage manufacturing industry continues to rely on fragmented tools such as spreadsheets, email, phone calls, and disconnected software systems to coordinate production activities. As companies expand, these manual processes often lead to communication delays, approval bottlenecks, and operational inefficiencies. The growing scale of the industry highlights the need for better solutions, with the global beverage packaging market valued at approximately \$157.7 billion in 2023 and projected to reach \$222.1 billion by 2030 (Grand View Research, 2024). This increasing complexity has created a demand for centralized, scalable information systems that improve collaboration among suppliers, production teams, logistics providers, and clients (Zimmerman, 2026).

1.2 Problem and Needed Solution Characteristics

To address communication delays, fragmented data, and manual processes within beverage manufacturing, an effective solution must improve coordination and automate key portions of the production workflow. The following characteristics identify the capabilities needed to address these challenges.

1.2.1 Automated request and approval workflows

The solution should automate product requests and artwork approvals while allowing human intervention or escalation when full automation is not appropriate.

1.2.2 Automated quoting and negotiation support

The solution should automate quote calculations and partially automate the negotiation process while allowing users to intervene when necessary.

1.2.3 Automated purchasing and supplier coordination

The solution should support purchase order placement, supplier responses, and follow-up activities to reduce repetitive manual communication.

1.2.4 Integrated issue tracking and feedback

The solution should capture internal user feedback and connect it to issue tracking so emerging problems and workflow shortcomings can be identified and addressed.

1.2.5 Centralized operational coordination

The solution should reduce repetitive manual work, communication delays, approval bottlenecks, limited visibility, and fragmented data by coordinating production, logistics, finance, suppliers, and communication through a centralized platform. Figure 1 illustrates how these activities are coordinated throughout the proposed solution process.

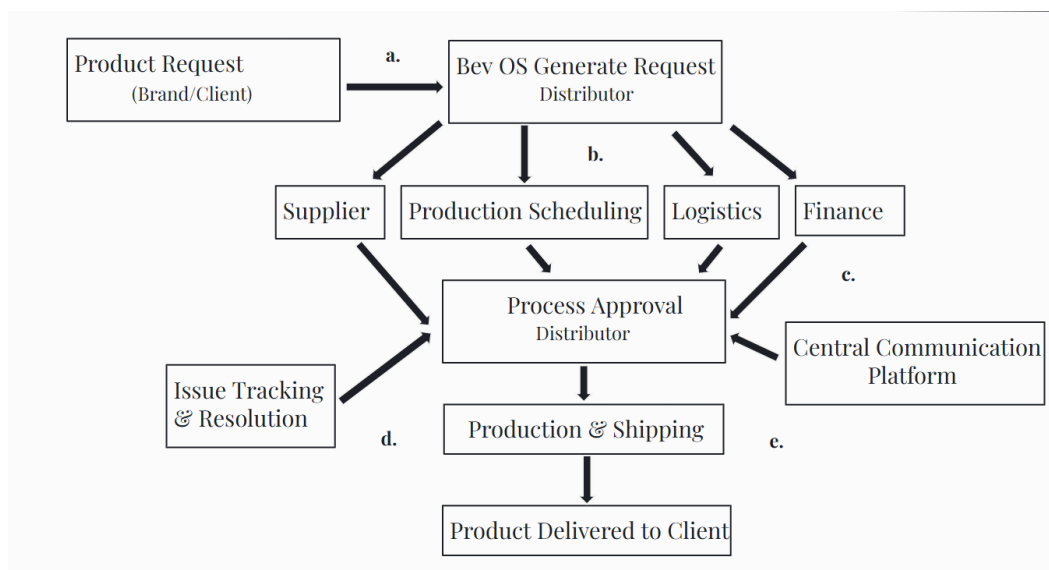


Figure 1: Solution Process Flow

1.3 Product Introduction

BevOS is a centralized workflow management platform designed for the beverage manufacturing industry. It connects brands, suppliers, distributors, co-packers, logistics providers, and internal teams through a single, integrated system. By replacing manual coordination with centralized workflows, BevOS improves communication, increases operational visibility, and streamlines production activities. A functional prototype will demonstrate how the system's interface, services, modules, and database work together to solve the industry's coordination challenges.

2 BevOS Product Description

BevOS is a centralized workflow management platform designed to improve coordination throughout the beverage manufacturing process. The system provides a single platform where brands, distributors, suppliers, and other authorized users can manage and monitor activities that would otherwise be handled through separate spreadsheets, emails, phone calls, and software systems. BevOS will support workflows involving product requests, quotes, approvals, production, logistics, documentation, communication, and reporting. Its primary goal is to create a single source of truth for operational information while reducing repetitive manual tasks, communication delays, approval bottlenecks, and fragmented data. By connecting these activities within one system, BevOS will provide users with greater visibility into the status of a product as it moves from the initial request through production and ultimately to delivery.

2.1 Key Product Features and Capabilities

BevOS provides several integrated capabilities that support the beverage manufacturing workflow. Instead of requiring users to manage individual stages of production through separate tools, BevOS brings these activities together within a common interface and shared data environment. Key capabilities include quote management, centralized communications,

production tracking, logistics coordination, document storage, and reporting. The system will also support automated and partially automated workflows for activities such as product requests, approvals, quote calculations, purchase order follow-up, and issue tracking.

An important capability of BevOS is the connection between these functions. Information created during one stage of the workflow can remain available as a product moves through subsequent stages, reducing the need to repeatedly transfer information between spreadsheets, emails, and other systems. Authorized users can access current information about requests, approvals, production status, shipping activities, documents, and other relevant workflow data. This centralized approach is intended to improve communication and operational visibility while reducing the delays and errors associated with fragmented information and repetitive manual processes.

2.2 Major Components (Hardware/Software)

BevOS will primarily be a software-based system and will not require specialized hardware for end users. Users will access the system through standard computers or other devices capable of running a modern web browser and connecting to the internet. The application itself will require server infrastructure to host the BevOS user interface, application services, and database. The specific production hosting hardware or cloud infrastructure has not yet been defined in the current design.

As shown in the Major Functional Components Diagram (MFCD, shown in Figure 2), BevOS is organized into multiple software layers. At the top are the system's users, including brands, distributors, and suppliers. These users interact with the system through the BevOS User Interface. The UI communicates with BevOS Core Services which provide the shared application

logic needed to process requests, enforce system rules, coordinate the functional modules, and manage the exchange of data throughout the system.

The core services connect to six primary functional modules: Quotes, Communications, Production, Logistics, Document Storage, and Reports. Each module is responsible for a specific portion of the beverage manufacturing workflow while remaining connected through the BevOS core services. A PostgreSQL relational database provides persistent data storage for the application. This architecture allows BevOS to centralize information while separating major system functions into manageable software components.

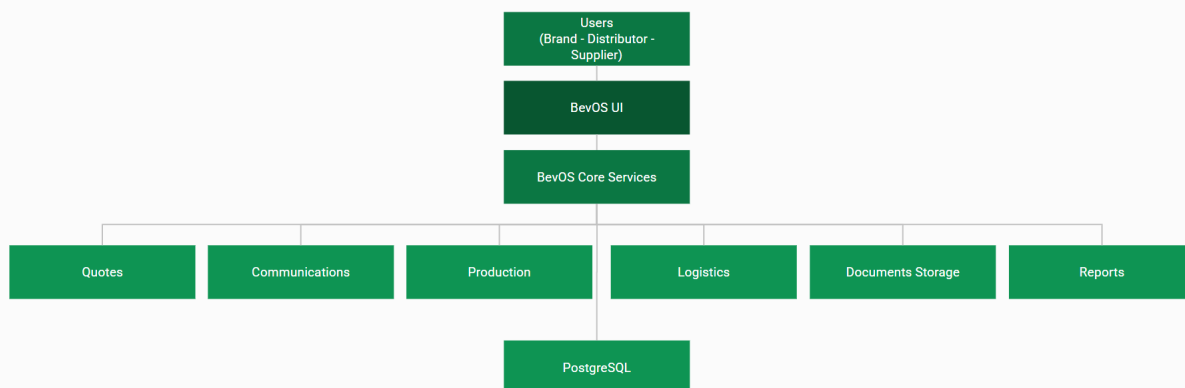


Figure 2: Major Functional Components Diagram

3 Identification of Case Study

BevOS is being developed for Nice Package Company, a company operating within the beverage manufacturing industry. The company coordinates activities among brands, suppliers, distributors, production teams, logistics providers, and other stakeholders throughout the manufacturing process. These activities currently rely on multiple forms of communication and disconnected tools, which can result in communication delays, fragmented information, repetitive manual tasks, and difficulty tracking the status of production activities.

BevOS is intended to address these challenges by providing Nice Package Company with a centralized platform for managing product requests, quotes, approvals, production, logistics, documents, communications, and reporting. By bringing these activities together, the system will provide users with a more consistent view of information as a product moves from the initial request through production and delivery.

Although BevOS is being developed around the needs of Nice Package Company, the underlying workflow challenges are not necessarily unique to a single organization. In the future, the system could potentially be adapted for use by other beverage manufacturers, co-packers, distributors, suppliers, and organizations that manage similar production and fulfillment workflows.

4 Glossary

- **Approval Workflow** - A process used to review and approve operational tasks before they move to the next stage.
- **BevOS** - A centralized workflow-management platform designed for beverage manufacturing operations.
- **Client** - An organization or individual purchasing services or products from another participant in the manufacturing process.
- **Co-Packer** - A third-party company that manufactures and packages beverages for another company.
- **Core Services** - The shared BevOS system software layer that processes requests, applies system rules and coordination functional modules, and manages data exchange.
- **Dashboard** - A visual interface displaying operational information and performance metrics.
- **Distributor** - An organization responsible for delivering products from manufacturers to retailers or customers.
- **ERP (Enterprise Resource Planning)** - Software that integrates core business processes into a unified system.
- **Logistics** - The planning and coordination of transportation and delivery activities.
- **MFCD (Major Functional Components Diagram)** - A diagram illustrating the primary hardware and software components of the system.
- **Nice Package Company** - The beverage-industry client and domain partner for whom BevOS is being developed.
- **PostgreSQL** - An open-source relational database management system used to store application data.
- **Role-Based Access Control** - A security method that grants system permissions according to the user's assigned role and responsibilities.
- **Single Source of Truth** - A trusted, centralized set of current information used consistently by authorized stakeholders.

- **Stakeholder** - A person or organization that participates in or is affected by a beverage-manufacturing workflow.
- **Workflow** - A sequence of business activities completed to achieve a specific objective.

5 References

- Grand View Research. (2024). Beverage Packaging Market Size, Share & Trends Analysis Report.
- Kennedy, T. (2026). CS410 Professional Workforce Development.
- Wilson, T. (2026). BevOS Presentation / Nice Package Co. Guest Presentation.
- Zimmerman, S. (2026) How the food and beverage industry is rethinking manufacturing investments. Food Dive.